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Native fishes in the Truckee River: Are in-stream structures and patterns of population genetic structure related?

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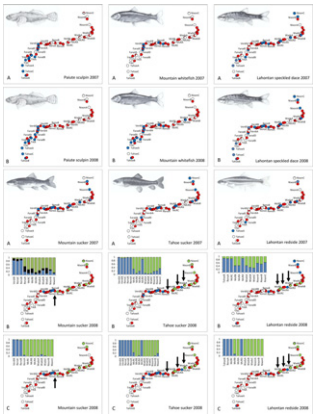
HIGHLIGHTS

- Impact of in-stream structures assessed for a suite of native fishes.
- Microsatellite markers and Bayesian genotype clustering were applied.
- Population subdivision was not observed a year after high flow conditions.
- Population subdivision was observed during low flow conditions for three species.
- Impacts to movement by in-stream structures can be mitigated by flow management.

GRAPHICAL ABSTRACT



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ABSTRACT

In-stream structures are recognized as significant impediments to movement for freshwater fishes. Apex predators such as salmonids have been the focus of much research on the impacts of such barriers to population dynamics and population viability however much less research has focused on native fishes, where in-stream structures may have a greater impact on long term population viability of these smaller, less mobile species. Patterns of genetic structure on a riverscape can provide information on which structures represent real barriers to movement for fish species and under what specific flow conditions. Here we characterize the impact of 41 dam and diversion structures on movement dynamics under varying flow conditions for a suite of six native fishes found in the Truckee River of California and Nevada. Microsatellite loci were used to estimate total allelic diversity, effective population size and assess genetic population structure. Although there is spatial overlap among species within the river there are clear differences in species distributions within the watershed. Observed population genetic structure was associated with in-stream structures, but only under low flow conditions. High total discharge in 2006 allowed fish to move over potential barriers resulting in no observed population genetic

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structure for any species in 2007. The efficacy of in-stream structures to impede movement and isolate fish emerged only after multiple years of low flow conditions. Our results suggest that restricted movement of fish species, as a result of in-stream barriers, can be mitigated by flow management. However, as flow dynamics are likely to be altered under global climate change, fragmentation due to barriers could isolate stream fishes into small subpopulations susceptible to both demographic losses and losses of genetic variation.

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1. Introduction

Many freshwater fish species are highly impacted by anthropogenic mediated alterations to river ecosystems (Rowe et al., 2009; Strayer, 2010; Branco et al., 2012, Matono et al., 2013; Van Haverbeke et al., 2013; Walters et al., 2014). Not the least of these are dams and water diversions for municipal and agriculture use, which fragment large flowing systems into smaller isolated reaches reducing habitat size, habitat diversity, and fish populations. Smaller less diverse habitats limit the ability of individuals to fulfill all life history requirements, thus negatively affecting population viability (Alexiades et al., 2012; Cooney and Kwak, 2013; Keefer et al., 2013; Santos et al., 2013; Van Haverbeke et al., 2013). In many instances anthropogenic impacts have resulted in listings under the United States Endangered Species Act (ESA) or the International Union for Conservation for Nature (IUCN). For most species, ESA listing has been concomitant with losses of genetic variation, and thus, evolutionary potential (Garrigan et al., 2002; Hurt and Hedrick, 2004; Dowling et al., 2012; Peacock and Dochtermann, 2012). As the impacts of global climate change intensify, human pressure on water resources, for example, in the western United States will undoubtedly increase, escalating both conservation concerns and challenges to maintain functioning aquatic ecosystems, their native species, and the ecosystem services they provide.

Native fishes that are not of commercial or recreational interest have few data to assess their status in large parts of their ranges precluding informed decisions regarding extinction vulnerability (Cooke et al., 2005; Quist et al., 2006). However, the ESA single species listing approach may ignore important ecosystem processes critical to population persistence for suites of interacting species (Franklin, 1993; Orians, 1993; Franklin, 1994; Tracy and Brussard, 1994; Whiteley et al., 2006). A more "ecosystem based" approach would be to assess the ecological viability of multiple species with multiple potential environmental stressors with the aim of managing for properly functioning ecosystems that support biotic communities (Kviberg and Craig, 2006; Edgar et al., 2016). Recent examples of an ecosystem based approach include Teichert et al.'s (2016) work on the impact of nine anthropogenic stressor categories on 90 estuary fisheries in the Northeast Atlantic and McDonald-Madden et al.'s (2016) use of food-web theory to guide ecosystem management of species networks.

Here we focus on the movement dynamics of a suite of native fishes found in the Truckee River in western Nevada and eastern California in the face of anthropogenic barriers. As with many watersheds in western North America, the Truckee River has been highly impacted by anthropogenic perturbations over the 20th and 21st centuries. A series of in-stream dam structures and water diversions on the river impound water for the urban centers of Reno and Sparks, Nevada and provide water for irrigation. Construction of a large in-stream dam diversion as part of the Newlands Reclamation Act of 1902 (P.L. 57-161) in 1905 and channelization by the Army Corp of Engineers in the late 1960′s greatly impacted flow dynamics, reduced recruitment in riparian cottonwoods stands in the lower river, and altered invertebrate community structure (USFWS Truckee River Recovery Implementation Team, 2003).

Despite the fact that the native fish species found in the Truckee River watershed are fairly common throughout their respective ranges, there is little known about their population ecology in general and almost no information available for populations found within the Truckee River watershed specifically (Baker, 1967; Vondracek et al., 1982; Marrin et al., 1984; Johnson, 1984; Vinyard and Yuan, 1996; Light, 2005; Dauwalter and Rahel, 2008; Taylor et al., 2012; Benjamin et al., 2014). Therefore the impact of anthropogenic disturbance, including the role that in-stream structures play in restricting movement, on the long-term population viability for many of these species, which range in size from small cyprinids, Lahontan speckled dace (*Rhinichthys osculus robustus*), Lahontan redside (*Richardsonius egregious*) and cottids, Paiute sculpin (*Cottus beldingi*) to larger catostomids, mountain sucker (*Catostomus platyrhynchus*), Tahoe sucker (*Catostomus tahoensis*) and salmonids, mountain whitefish (*Prosopium williamsoni*) remains unknown.

In this study we characterize the spatial distribution of the six fluvial, native fish species found in the main stem Truckee River from its origin at Lake Tahoe, California to its terminus at Pyramid Lake, Nevada (Fig. 1). Hypotheses concerning both flow dynamics, which are highly variable both temporally and seasonally in this watershed, and the presence and type of potential barriers on movement dynamics of these fishes were tested. All fish species are not uniformly distributed throughout the river due to differing ecological requirements such as water temperature, flow and foraging needs. Additionally the type of in-stream structure varies throughout the watershed. All in-stream structures were recently evaluated for fish passage by United States Fish and Wildlife Service (USFWS) (Table 1), and these designations were used to generate predictions regarding potential barriers and individual species.

Patterns of fish genetic structure in a riverscape can provide information with which to assess potential barriers to movement and the impact of flow conditions. We predict that population substructure will be present for those species whose distributions overlap with structures classified as complete in-stream barriers. Additionally, temporal and seasonal barriers are likely to impact smaller, less mobile fishes such as Paiute sculpin, Lahontan redside, and Lahontan speckled dace, during protracted low flow conditions leading to additional population substructure and small isolated groups of individuals. Smaller fishes are likely to be impacted disproportionally by seasonal and temporal barriers as studies on movement dynamics show that fish length can play an important role in determining home range size and movement patterns. Although none of the species from this study were included in a meta-analysis of dispersal patterns in fluvial fishes, Radinger and Wolter (2014) found fish length to be positively correlated with movement distances. The cottids had among the shortest movement distances and a narrower total range of distances moved (1–100 m), followed by cyprinids, salmonids, and catostomids (Radinger and Wolter, 2014).

Because many more barriers classified as seasonal and temporal are found on the Truckee River, they have the potential to impact movement dynamics for the smaller fish species whose distributions largely overlap with them. Only during periods of high flow could fish move downstream over complete in-stream barriers, but larger fish species may be able to negotiate other in-stream structures during low flows thereby reducing the impact of such structures on movement, gene flow, and population size. Barrier height can also be an impediment to movement especially for smaller fish species. In-stream structures with perch height > 15 cm represented barriers to upstream movement for the prickly (*Cottus asper*) and coastrange (*Cottus aleuticus*) sculpins both small species (70–300 mm; 60–170 mm respectively) (LeMoine and Bodensteiner, 2014). Small populations of fish isolated between temporal and seasonal barriers are more susceptible to demographic

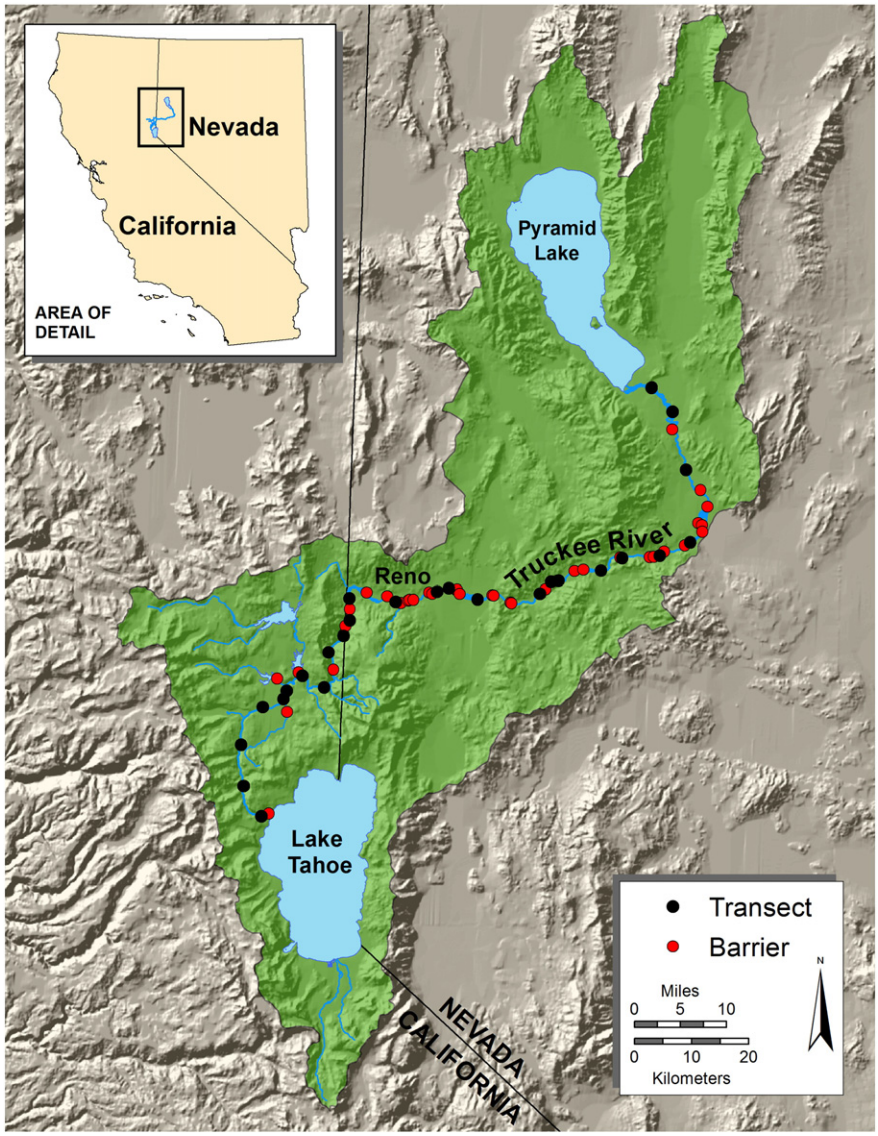


Fig. 1. The Truckee River watershed in eastern California and western Nevada. Sampling transects and barriers are indicated by black and red circles respectively.

and genetic stochasticity, which can result in losses of genetic variation and increased risk of local extirpation, especially if low flows persist for long periods of time, e.g. under drought conditions. Therefore we predict that species whose distributions overlap multiple temporal and seasonal barriers will have smaller genetic effective population sizes, reflective of more frequent, but transitory, population genetic structure, which emerges during periods of low flow. Here we use microsatellite loci and landscape genetic approaches to tease apart the effects of barrier type and flow conditions on movement patterns, maintenance of genetic variation and effective population size for fish species whose distributions are interspersed with structures deemed fish passage barriers.

2. Materials and methods

2.1. Study species

General characteristics of study species including life span, adult size range and habitat are listed in Table 2. Mountain suckers are widely distributed in Western North America and are often found in association with Tahoe suckers and Lahontan speckled dace in Nevada and California (Decker and Erman, 1992). The Tahoe sucker is endemic to the hydrographic Lahontan basin in northern Nevada, northeastern California and southeastern Oregon. Mountain and Tahoe suckers co-occur in the Lake Tahoe basin and the Truckee River watershed (Moyle, 1995) and have experienced declines in recent years. The mountain sucker is currently listed as a species of special concern in California (CDFG, 1995).

Most of what is known about movement dynamics among the catostomids comes from larger species where both resident sedentary and mobile individuals have been documented within populations. Tagged white suckers (*Catostomus commersonii*) in a main stem and tributary system revealed both sedentary resident and a small number of highly mobile individuals (Bliss et al., 2015). Home range size for the Sonora (*Catostomus insignis*) (200–800 mm) and desert (*Catostomus clarkii*) (100–410 mm) suckers in the Gila River, a smaller tributary to the Colorado River, were small ≥ 150 m, but desert suckers moved over 600 m in the Sonoita Creek in Arizona (Booth and Shipley, 2012; Booth et al., 2014).

The Paiute sculpin is the only sculpin species found in the Lahontan hydrographic basin (Sigler and Sigler, 1987; Sigler and Sigler, 1996). Though not listed under the United States ESA, Paiute sculpin co-occur with species that are listed, including the Lahontan cutthroat trout (*Oncorhyncus clarkii henshawi*), cui-ui (*Chasmistes cujus*), and bull

Table 1
List of all transects (T) per section (S) (bolded) and potential barriers with per site elevation, latitude and longitude and USFWS barrier assessment (sections: S8 = Tahoe, S7 = Truckee, S6 = Farad, S5 = Verdi, S4 = Reno, S3 = McCarran, S2 = Wadsworth, S1 = Nixon).

Transect/barrier name	Section (S) Transect (T)	Elevation (m)	Latitude	Longitude	USFWS assessment of potential in-stream fish passage barriers
1. Lake Tahoe Dam	Barrier	6256	39.167	−120.144	In-stream dam complete barrier
TahoeB	**S8 T2**	**6225**	**39.162**	**−120.159**	
TahoeC	**S8 T3**	**6079**	**39.208**	**−120.198**	
TahoeA	**S8 T1**	**5909**	**39.274**	**−120.206**	
TruckeeC (TrkC)	**S7 T3**	**5751**	**39.334**	**−120.164**	
2. Prosser Creek Dam	Barrier	5737	39.380	−120.138	Off channel on Prosser Creek
TruckeeB (TrkB)	**S7 T2**	**5676**	**39.348**	**−120.123**	
TruckeeA (TrkA)	**S7 T1**	**5598**	**39.361**	**−120.117**	
3. Martis Creek Dam	Barrier	5825	39.327	−120.115	Off channel on Martis Creek
4. Boca Dam	Barrier	5565	39.390	−120.094	Off channel Little Truckee River
FaradA	**S6 T1**	**5472**	**39.386**	**−120.086**	
FaradB	**S6 T2**	**5400**	**39.368**	**−120.041**	
5. Farad Power Diversion	Barrier	5290	39.397	−120.024	In-stream dam complete barrier
FaradD	**S6 T4**	**5169**	**39.423**	**−120.035**	
FaradC	**S6 T3**	**5076**	**39.450**	**−120.005**	
6. Fleish Power Diversion	Barrier	5074	39.452	−120.006	In-stream dam complete barrier
7. Steamboat Ditch Diversion	Barrier	5017	39.466	−120.003	Temporal and seasonal barrier
VerdiA	**S5 T1**	**4984**	**39.475**	**−119.994**	
8. Washoe Power Diversion and Highland Ditch	Barrier	4780	39.520	−119.962	In-stream dam complete barrier
9. Last Chance Ditch Diversion	Barrier	4658	39.515	−119.92	Not a barrier to fish passage
VerdiC	**S5 T3**	**4854**	**39.509**	**−119.996**	
10. Verdi Power Diversion and Coldron Ditch	Barrier	4917	39.493	−119.995	In-stream dam complete barrier
VerdiB	**S5 T2**	**4637**	**39.507**	**−119.902**	
11. Lake Ditch Diversion	Barrier	4612	39.505	−119.891	Not a barrier to fish passage
12. Southside Ditch Diversion	Barrier	4611	39.509	−119.876	Not a barrier to fish passage
13. Orr Ditch Diversion	Barrier	4593	39.510	−119.874	Not a barrier to fish passage
14. Chalk Bluff Pump Station	Barrier	4582	39.510	−119.866	Fish ladder present
15. Idlewild Pond Diversion	Barrier	4515	39.523	−119.835	Not a barrier to fish passage
16. Idlewild Pond Return Drain	Barrier	4497	39.520	−119.828	Not a barrier to fish passage
RenoC	**S4 T3**	**4500**	**39.524**	**−119.819**	
17. Cochran Ditch Diversion	Barrier	4507	39.524	−119.817	Not a barrier to fish passage
18. Wingfield Park Dams	Barrier	4504	39.525	−119.817	Temporal and seasonal barrier
RenoB	**S4 T2**	**4448**	**39.530**	**−119.795**	
19. Eastman Ditch Diversion	Barrier	4481	39.527	−119.778	Temporal and seasonal barrier
20. Glendale Ditch Diversion	Barrier	4480	39.528	−119.778	Temporal and seasonal barrier
21. Pioneer Ditch Diversion	Barrier	4406	39.522	−119.773	Temporal and seasonal barrier
RenoA	**S4 T1**	**4394**	**39.514**	**−119.736**	
22. North Truckee Drain confluence	Barrier	4400	39.521	−119.706	Not a barrier to fish passage
23. Steamboat Creek and Sparks-Reno Sewage Plant	Barrier	4360	39.521	−119.703	Potential water quality barrier
24. Lagomarsino Noce Ditch Diversion	Barrier	4368	39.511	−119.669	Not a barrier to fish passage
25. Murphy Ditch Diversion	Barrier	4370	39.510	−119.667	Not a barrier to fish passage
McCarran RanchC (McRC)	**S3 T3**	**4309**	**39.526**	**−119.61**	
26. McCarran Ditch Diversion	Barrier	4290	39.533	−119.599	Not a barrier to fish passage
McCarran RanchB (McRB)	**S3 T2**	**4284**	**39.545**	**−119.587**	
McCarran RanchA (McRA)	**S3 T1**	**4277**	**39.547**	**−119.573**	
27. Hill Ditch Diversion	Barrier	4260	39.563	−119.541	Temporal and seasonal barrier
28. Tracy Power Plant	Barrier	4246	39.565	−119.522	Not a barrier to fish passage
McCarran RanchD (McRD)	**S3 T4**	**4202**	**39.565**	**−119.487**	
29. Derby Dam and TCID Diversion	Barrier	4212	39.586	−119.448	In-stream dam complete barrier
WadsworthA (WadsA)	**S2 T1**	**4189**	**39.585**	**−119.444**	
30. Marble Bluff Dam	Barrier	3844	39.856	−119.393	In-stream dam complete barrier
31. Washburn Ditch Diversion	Barrier	4129	39.588	−119.388	Temporal and seasonal barrier
32. Outlet Ditch Christensen Ranch Pond	Barrier	4155	39.589	−119.38	Not a barrier to fish passage
WadsworthB (WadsB)	**S2 T2**	**4132**	**39.591**	**−119.368**	
33. Gregory Ditch Diversion	Barrier	4146	39.597	−119.359	Temporal and seasonal barrier
34. Herman Ditch Diversion	Barrier	4105	39.608	−119.318	Temporal and seasonal barrier
WadsworthC (WadsC)	**S2 T3**	**4099**	**39.613**	**−119.306**	
35. Pierson Ditch Diversion	Barrier	4071	39.613	−119.306	Not a barrier to fish passage
36. Cersola Ditch Diversion	Barrier	4045	39.643	−119.291	Temporal and seasonal barrier
37. Olinghouse No. 3 pump	Barrier	3979	39.696	−119.288	Not a barrier to fish passage
38. Olinghouse No. 1 Pump	Barrier	4050	39.640	−119.284	Not a barrier to fish passage
39. Proctor Ditch Diversion	Barrier	4062	39.630	−119.283	Temporal and seasonal barrier
40. Gardella Ditch Diversion	Barrier	4006	39.669	−119.274	Temporal and seasonal barrier
NixonA	**S1 T1**	**3988**	**39.727**	**−119.319**	
41. Numana Dam and Pyramid Indian Diversion	Barrier	3916	39.790	−119.35	In-stream dam complete barrier
NixonB	**S1 T2**	**3903**	**39.818**	**−119.35**	
NixonC	**S1 T3**	**3850**	**39.854**	**−119.394**	

trout (*Salvelinus confluentus*). Despite that in general sculpin are thought to be relatively sedentary there is considerable variability in movement patterns both within and among cottid species (Bailey, 1952; Hill and Grossman, 1987; Fischer and Kummer, 2000; Natsumeda, 2007). Distance moved by slimy sculpin (*Cottus cognatus*) individuals varied both among watersheds and age classes with younger fish moving father than older fish (Clarke et al., 2015). Twenty percent of marked mottled sculpin (*Cottus bairdii*) moved >100 m with a

Table 2
Study species, life span, adult size range and habitat.

Family	Species	Life span	Adult size range (mm)	Habitat	References
Catostomidae	Mountain sucker	≤5 years	100–230	Favor pools with ample cover in the form of logs, overhanging banks, and aquatic vegetation	Sigler and Sigler (1987, 1996), Decker and Erman (1992)
	Tahoe sucker	≤5 years	≤150	Benthic fauna found in cool waters (lakes and streams) deeper waters in the daytime and near shore shallow waters at night	Sigler and Sigler (1987, 1996), Decker and Erman (1992)
Cottidae	Paiute sculpin	3–5 years	≤90	Benthic fauna and can be found in a variety of substrate types including silt, under logs, gravel, boulders, and in aquatic vegetation, but prefer coarse substrates	Dietsch (1950), Ebert and Summerfelt (1969), Sigler and Sigler (1987, 1996)
Cyprinidae	Lahontan speckled dace	≤3 years	50–63.5	Shallow, swift running, cold riffles	La Rivers (1962), Sigler and Sigler (1987, 1996)
	Lahontan redside	≤4 years	25–76	Most common in pools and slow runs, and littoral zones in lakes	La Rivers (1962), Sigler and Sigler (1987, 1996)
Salmonidae	Mountain whitefish	7–8 years	260–440	Tend to be found in pool habitat	Sigler and Sigler (1987, 1996), Whiteley et al. (2006)

single fish moving close to a kilometer in response to in-stream barrier removal (Deboer et al., 2015).

The Lahontan speckled dace is a forage fish commonly preyed upon by salmonids and other game fish and is endemic to the Lahontan hydrographic basin (La Rivers, 1962; Sigler and Sigler, 1987). Although tolerant of environmental variation within their respective habitats, speckled dace subspecies are sensitive to degradations in water quality (Carter and Hubert, 1995; Robinson et al., 1998; Robinson and Childs, 2001; Propst and Gido, 2004). There have been no systematic studies on the population ecology of this particular subspecies anywhere within its range including the Truckee River watershed.

The Lahontan redside is endemic to interior basins of northern California and northern and western Nevada including the Truckee, Carson, Walker, Susan, Quinn, Humboldt and Reese river systems (Sigler and Sigler, 1987). They are also found in the saline terminal lakes in Nevada, Walker, and Pyramid lakes, as well as oligotrophic Lake Tahoe. This cyprinid is an important forage species for piscivorous game fish. There is a large size range among cyprinid species, but species in the *Rhinichthys* (N = 7 extant species) and *Richardsonius* (N = 2) genera tend to be small (60–225 mm; 76–120 mm respectively). Although very little information is available on movement dynamics for species in these genera, a single study in Sagehen Creek, a tributary which historically flowed into the Truckee River and now flows into a reservoir, with the same complement of native fishes as the main stem river, found that changes in seasonal abundance of Lahontan redsides was likely tied to breeding migrations of 500–1000 m from the reservoir into Sagehen creek (Decker and Erman, 1992). However, this study did not quantify movement patterns for stream resident fish (Decker and Erman, 1992). Home range data for other *Rhinichthys*, longnose (*Rhinichthys cataractae*) and blacknose dace (*Rhinichthys atratulus*) from a small Applacachian stream, show the majority of individuals moved <500 m, with a few larger and heavier individuals, 0.2 and 0.3% of all longnose and blacknose dace captured, moving farther distances (Larson et al., 2002).

Mountain whitefish are widely distributed throughout the northwestern United States where they are found in most major river basins, but not in the interior Great Basin. Populations of mountain whitefish in the Sierra Nevada are highly disjunct from the rest of the species distribution (Sigler and Sigler, 1987; Whiteley et al., 2006). Where they have been studied mountain whitefish have broad spatio-temporal movements within large river systems, which suggest they require habitats that are widely spaced in order to acquire sufficient food resources to sustain metabolic needs (Al-Chokhachy and Budy, 2008; Benjamin et al., 2014). As with many other fish species populations are primarily comprised of age correlated resident and migratory individuals that move 100s of km. Connectivity of large river networks appear to be important for long term population persistence (Benjamin et al., 2014).

2.2. Study system

The Truckee River in eastern California and western Nevada is one of three major rivers which flow from the Sierra Nevada into the hydrographic Lahontan basin. The main stem Truckee River flows 195 km from oligotrophic Lake Tahoe in California (elevation 1900 m) to endorheic Pyramid Lake (elevation 1157 m) in Nevada through both high alpine and desert ecoregions (Fig. 1). The portion of the Truckee River watershed in California has a temperate semiarid climate characterized by cold wet winters and warm dry summers, whereas the Nevada portion of the watershed lies mostly in the rain shadow of the Sierra Nevada and has an arid desert climate with much less precipitation and hotter summers. The different climatic conditions in the watershed are apparent in precipitation data – Truckee, CA receives 78 cm mean annual precipitation, while Reno, NV only receives an average of 18 cm annually (National Oceanic and Atmospheric Administration, NOAA). The differences in climate, geology, and elevation create different environmental conditions which support different faunal and floral communities within the watershed. As with many rivers in the western United States, the flow is highly regulated on the Truckee through four major reservoir impoundments and at least 41 in-stream structures or irrigation ditch diversions.

2.3. Sampling

The river was divided into 8 large continuous sampling sections (Tahoe, Truckee, Farad, Verdi, Reno, McCarran, Wadsworth and Nixon; Table 1). Within each section three to four transects were established depending on the length of the section, and the number of available habitat types contained therein. Sampling transects were 500 m long and incorporated as many in-stream habitat types as possible. In addition to the 500 m transects a 100 m stretch within each 500 m transect was identified for sampling of shoreline habitat. The beginning and end points of each transect were delineated using hand held GPS units. Transects were sampled twice a year in summer and fall in order to account for seasonal environmental conditions and both resident and migratory life histories.

Raft, barge, and backpack electrofishing methodologies were employed to ensure that all species were adequately surveyed. Sampling was conducted May through October in 2007 and 2008. Fish within the main channel were sampled using a raft electrofisher once per month during May–August, after August flows are typically too low to permit raft sampling. In wadeable transects with discharges <-225 cubic feet per second (cfs) a smaller cataraft towed barge was used with the raft electrofishing equipment mounted on the cataraft. The barge was placed at the bottom of the 500 m transect and pushed upstream, sampling as much of the river as possible. Because float

shockers are most effective with the larger species, smaller fish species and younger age classes of larger species were sampled using a backpack electrofisher sampling protocol along the shoreline from May–October, as river flows permitted.

At each sampling location the species, individual size (fork length, mm) and weight (g) were recorded for each individual caught. All individuals, >200 mm, received an individually numbered floy tag. Fin clips were collected from all adults identified by length (mm) for genetic analyses and were placed in wax paper, dried, and stored in individual coin envelopes labeled with species identified, date collected and sampling location. All fish sampled were released back into the river at their capture location.

2.4. Water discharge

Monthly flow (m^3/s) and total discharge (m^3) data collected from 1945 to 2013 for eight gauging stations on the Truckee River were obtained from the United States Geological Survey (USGS; http://wdr.water.usgs.gov/wy2006/search.jsp) in order to characterize stream flow dynamics in this system (Fig. 2). Discharge data for 2002–2008 were compared in order to test for an association among high flow (2006) and low flow (2007, 2008) conditions with the distribution of the species, presence of potential barriers, and the spatial distribution of genetic variation.

2.5. In-stream structures

Of the 41 in-stream structures and irrigation ditch diversions on the main stem river, eight are large in-stream dam structures for generation of hydroelectric power and are considered complete in-stream barriers to upstream, but not downstream movement by USFWS (Table 1). Five complete in-stream barriers are in the upper watershed clustered between the FaradB and VerdiB transects (Fig. 3). Upstream from FaradB and above the FaradA transect the Truckee River has few main channel barriers, with the exception of the Lake Tahoe Dam at the outflow of the Truckee River, the three other barriers in this portion of the watershed are associated with reservoirs on tributaries that historically flowed into the Truckee River. The remaining three complete in-stream barriers are in the lower river below the Reno/Sparks metropolitan area and the McRD transect (Fig. 3). Twelve additional structures are considered temporal and seasonal barriers in the main part of the river between the FaradC and NixonA transects. The remaining structures are primarily ditch diversions and are not considered barriers to fish passage, however, could potentially be barriers when flows are very low. Steamboat Creek, below the RenoA transect, which drains into the main river channel receives treated sewage from the Sparks-Reno sewage plant and mercury from legacy mining, may be a water quality barrier to fish passage.

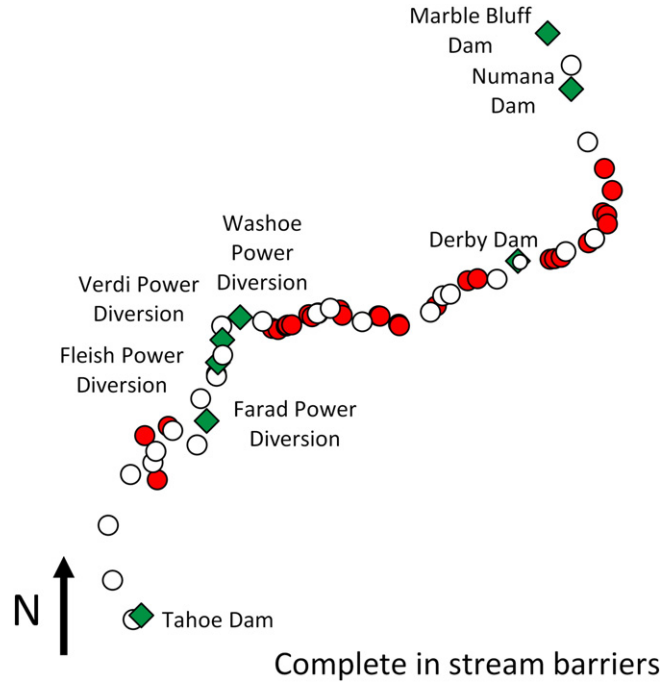


Fig. 3. Map of all potential barriers (red circles) and the 8 complete in-stream barriers (green diamonds) with names labeled. Sampling transects are represented by open circles.

2.6. Laboratory methods

Total genomic DNA was isolated using DNeasy96 Blood and Tissue kits (QIAGEN) according to the manufacturer's protocol and quantified per individual using a Labsystems Fluoroskan Ascent fluorometer. Neutral, nuclear di, tri, and tetra-nucleotide repeat microsatellite markers available for these species were used, or markers we optimized from closely related species (Table S1). Polymerase chain reaction conditions can be found in Supplementary documents (Table S2). PCR products were diluted to an appropriate concentration determined by dilution tests. 1 μl of diluted PCR product was added to 19 μl of HiDye Formamide/LIZ500 size standard (Applied Biosystems). Fragment analysis was carried out on an Applied Biosystems 3730 Genetic Analyzer at the University of Nevada, Nevada Genomics Center (http://www.ag.unr.edu/

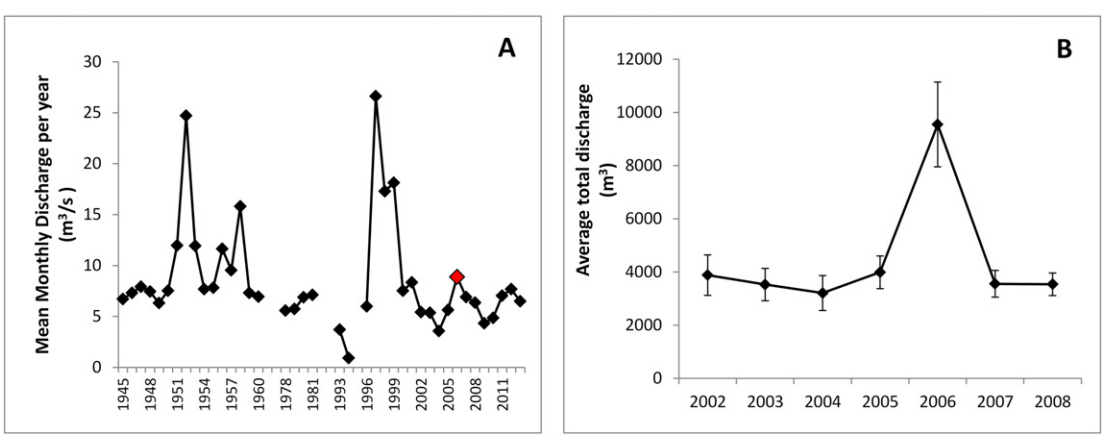


Fig. 2. Water discharge in the Truckee River. (A) Mean monthly discharge per year (m^3/s) in the Truckee River from 1945 to 2013. (B) Average total discharge (m^3) for 2002–2008 for multiple USGS gauging stations on the Truckee River. Data obtained from http://wdr.water.usgs.gov/wy2006/search.jsp.

genomics/), and all alleles generated were scored and binned using the ABI GeneMapper software (version 3.7).

2.7. Characterization of genetic variation

Genetic data were analyzed separately for each year and species. This was done to detect temporal trends in population genetic structure, and to look for patterns in the spatial dispersion of allelic variation in relation to water discharge and in-stream structures. The presence of null alleles and allelic dropout were tested for all loci using the program Micro-Checker (version 2.2.3; Van Oosterhout et al., 2004). Deviations from Hardy Weinberg equilibrium were assessed (HWE; F_{IS}) using the program FSTAT (version 2.9.3.2; Goudet, 2001). Any locus which showed systematic patterns of deviation from HWE or had evidence of null alleles or allelic dropout across all sampling locations was removed from the analysis. FSTAT was used to calculate the number of alleles (A) and allelic richness (R_S) per locus. Each species was characterized for observed (Ho) and expected (He) heterozygosity per locus per sampling location and averaged over all loci per sampling location using Microsatellite Toolkit in Excel (Park, 2001).

2.8. Population genetic structure

Pairwise F_{ST} among all sampling locations per species per year was calculated using the program FSTAT [version 2.9.3.2; Goudet, 2001). AMOVAs were conducted in GenAlEx (6.5; Peakall and Smouse, 2006; Peakall and Smouse, 2012) for each species for each year to characterize the partitioning of genetic variation on the landscape. Relatedness (r) among individuals within each transect in each year was calculated using the Lynch and Ritland (1999) method in GenAlEx to ensure that full sibling family groups were not being sampled.

Recent comparisons among multiple Bayesian clustering techniques suggests that datasets should be analyzed using multiple methods and all clustering patterns that do not have a biologically meaningful explanation should be suspect (Frantz et al., 2009). Contemporary clusters of genotypically similar adults for each species in each year were therefore analyzed using two Bayesian genotype clustering methods (STRUCTURE version 2.3.4, Pritchard et al., 2000 and BAPS version 5.2, Corander et al., 2008). In STRUCTURE, an admixture model was used wherein individuals with novel genotypes can be identified and assigned. A specified range of 10,000 iteration burn-in followed by five 500,000 Markov chain Monte Carlo (MCMC) replicates per k for $k = 1$–20 was used to approximate posterior allelic distributions against which individual genotypes were compared and assigned to a cluster. The ΔK method of Evanno et al. (2005) was applied to determine the optimal number of genotype clusters. In BAPs 10,000 input iterations were specified for admixture analysis and 10,000 input iterations were run specifying both 10 and 50 reference individuals per sampling location. Five replicates per k for $k = 1$–20 were conducted. Multiple Mantel tests were conducted in GenAlEx 6.5: 1) transects within each genotype cluster, and 2) transects between clusters identified per species per year to test for an isolation-by-distance pattern.

2.9. Effective population size

Effective population size (Ne) was calculated for all genotype clusters identified for each species in each year using NeEstimator (version 1.3; Peel et al., 2004). We used the linkage disequilibrium module based on a single point sample. The moment-based temporal method of Waples (1989), which is available in NeEstimator, was not used as the linkage disequilibrium method has "greater precision" with microsatellite data (Waples and Do, 2010). Additionally, because different numbers of genotype clusters were observed per year per species pooling them to utilize the temporal method would be inappropriate.

3. Results

3.1. Species distribution

Five thousand four hundred and twenty-six individuals were sampled (all species: 2007, N = 1894; 2008, N = 3548; Table 3). Fin clip samples were collected and genotypes generated for 2643 adults (all species: 2007, N = 792; 2008, N = 1851). Although there was spatial overlap among species within the river there were clear differences in distribution within the watershed related to habitat characteristics such as stream flow, temperature, and gradient (Table 3, Figs. 3 and 4). Paiute sculpin were only found in the upper reaches of the river, in the VerdiB transect and above, whereas mountain whitefish were found in the upper and middle reaches primarily above the RenoA transect. Although in 2008, six mountain whitefish were found in the WadsA transect in the lower river. Both sucker species and Lahontan redside were found primarily in the lower sections of the river, in the Reno transects and below (Table 3). Only Lahontan speckled dace was found in all transects sampled.

3.2. Water discharge

USGS gauging station flow data from 1945 to 2013 show the highly variable nature of annual discharge in the Truckee River (Fig. 2a). Flows in 2006 were statistically higher compared to the previous four years (2002–2005) as well as flows recorded for 2007 and 2008 (Fig. 2b) ($F_{6,49} = 7.522$, $P = 0.000$, Tukey HSD multiple comparisons, $P = 0.000$). Whereas, 2002–2005, 2007, and 2008 were low flow years where flows were not statistically different among years ($F_{5,42} = 0.218$, $P = 0.953$, Tukey HSD multiple comparisons, $P \geq 0.968$; Fig. 2b).

3.3. Genetic variation

3.3.1. Catostomids

The mountain sucker had high levels of observed and expected heterozygosity as well as high allelic diversity at all loci (Table S3). There were no significant F_{IS} values in 2007, but significant F_{IS} for two loci in one transect in 2008 (VerdiB: US3, $F_{IS} = 0.246$; DLU451, $F_{IS} = 0.202$; $P = 0.00064$). The high positive F_{IS} values indicate a deficit of heterozygotes at those loci for fish sampled in that transect for that year, suggesting inbreeding a possible effect of subpopulation structure. AMOVA results indicate that 89–90% of molecular variance was within individuals, 9–10% among individuals, 0–1% among populations (transects), and 0–1% among regions (sampling sections i.e., Tahoe, Truckee, Farad, Verdi, Reno, McCarran, Wadsworth, and Nixon) for 2007 and 2008, respectively.

Tahoe sucker also had high levels of observed and expected heterozygosity and allelic diversity at all loci (Table S3). There were no significant F_{IS} values in 2007, but in 2008 there were significant F_{IS} values for single loci in five transects (RenoA, US9, $F_{IS} = 0.270$; McRA, US3, $F_{IS} = 0.306$; McRD, US9, $F_{IS} = 0.467$; NixonA, DLU451, $F_{IS} = 0.328$; NixonB, US3, $F_{IS} = 0.219$; $P = 0.0005$). The high positive F_{IS} values indicate a deficit of heterozygotes at those loci for fish sampled in those transects. AMOVA results indicate 78–84% of the molecular variance was within individuals, 14–21% among individuals, 1% among populations, and 0–1% among regions.

3.3.2. Cottids

Heterozygosity and allelic diversity per locus in the Paiute sculpin ranged from very low (LCE219, He = 0.206, Ho = 0.229; CGO310, A = 3) to very high (CGO18, He = 0.965, Ho = 0.926; A = 54; Table S3). There were no significant F_{IS} values in any year for any locus. The majority of molecular variance was within individuals (95%) with 4% among individuals, 0% among populations, and 1% among regions for 2007 and 2008.

Table 3
Total number of individuals (adults and juveniles) sampled per species, per transect, per year (2007, 2008). Dark gray shading indicates transects with the greatest abundance per species and light gray shading shows additional transects where species were found.

River Section	Paiute Sculpin		Mountain Whitefish		Mountain Sucker		Tahoe Sucker		Lahontan Redside		Lahontan Speckled dace	
	2007	2008	2007	2008	2007	2008	2007	2008	2007	2008	2007	2008
Tahoe	66	112	1	10	0	1	9	6	16	7	31	51
Truckee	69	189	1	14	0	1	0	0	0	0	23	15
Farad	63	170	4	19	4	3	1	0	0	0	7	10
Verdi	100	56	45	8	10	58	2	6	0	0	0	7
Reno	0	0	24	70	69	133	22	51	26	52	33	39
McCarran	0	0	1	0	85	70	237	485	256	401	16	61
Wadsworth	0	0	0	6	13	82	122	339	210	402	6	13
Nixon	0	0	0	0	40	13	96	118	98	154	88	306
Total	298	527	76	127	221	361	489	1005	606	1016	204	502

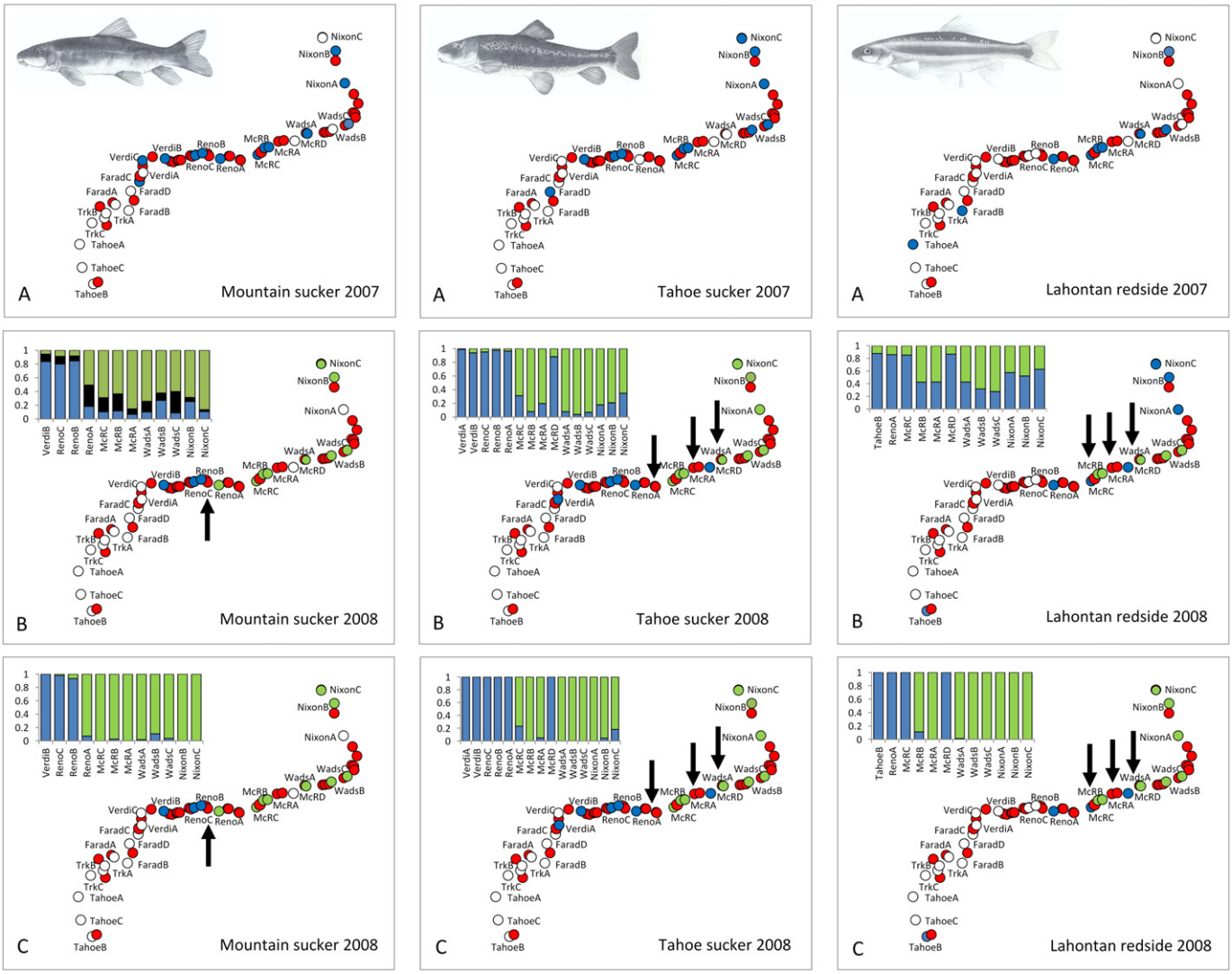


Fig. 4. Bayesian genotype clustering results. The sampling sites (open, blue or green circles) and barriers (red circles) are graphed by latitude (Y axis) and longitude (X axis) to show location on the river (see Fig. 1). The name of each transect is indicated (see Table 1). Each panel represents a single species and year. All transects where a species was found are indicated by a colored circle (blue or green). (A) Sampling locations for Mountain sucker, Tahoe sucker, and Lahontan redside in 2007 when no population genetic structure was observed; (B) STRUCTURE, and (C) BAPs results for species with population genetic structure in 2008. The color of the circle (blue or green) corresponds to the genotype cluster membership in each year of the study. Inset graphs represent the proportional membership per genotype cluster (Y axis) by transect (X axis). Arrows indicate structures which acted as barriers to movement for individuals species in 2008 as indicated by genotype cluster membership (see text for barrier details).

3.3.3. Cyprinids

Observed and expected heterozygosity and allelic diversity were high for all loci genotyped for Lahontan speckled dace (Table S3). There were no significant F_{IS} values in 2007 and a single significant F_{IS} at locus Lsou08 in the McRA transect in 2008 ($F_{IS} = 0.276$, $P = 0.0004$) indicating heterozygote deficiency. AMOVA results indicate 81–85% of the variance was within individuals, 13–18% among individuals, 0–1% among populations, and 0–2% among regions for 2007 and 2008 respectively.

Allelic diversity was high at all of the loci genotyped for Lahontan redside (Table S3). Observed and expected heterozygosity ranged from low for locus CypG1 (He = 0.367, Ho = 0.391) too high for locus CypG3 (He = 0.971, Ho = 0.872) and CypG27 (He, Ho = 0.967) (Table S3). There were no significant F_{IS} values in 2007, but significant positive F_{IS} for the CypG3 locus in the RenoA, McRB, McRA, McRD, WadsA transects ($F_{IS} = 0.232, 0.097, 0.174, 0.139, 0.158$; $P = 0.0008$) in 2008 indicating heterozygote deficits. AMOVA results show most of the molecular variance was within individuals 85–89% with 7–10% among individuals, 3–4% among populations, and 0–2% among regions for 2007 and 2008.

3.3.4. Salmonids

Mountain whitefish was the least abundant fish species in the Truckee River (2007, N = 76; 2008, N = 127; Table S3). Allelic diversity and heterozygosity were low for two loci COCL32 (He = 0.374, Ho = 0.327, A = 7) and OmyFGT25 (He = 0.161, Ho = 0.181, A = 3; Table S3). Although heterozygosity was relatively high for locus ONEμ8 (He = 0.572, Ho = 0.556) the number of alleles was low (A = 4) (Table S3). There were no significant F_{IS} values in either 2007 or 2008 ($P = 0.0007$). AMOVA results indicate 87–93% of the variance was within individuals, 6–10% among individuals, 0–2% among populations, and 1% among regions.

3.4. Population genetic structure

None of the six species sampled showed evidence of multiple genotype clusters in 2007, the year following the high flow year of 2006 (Fig. S1). While in 2008 multiple genotype clusters were observed for mountain sucker (STRUCTURE $k = 3$, $\Delta k = 121.76$; BAPs $k = 2$, $p = 0.961$), Tahoe sucker (STRUCTURE $k = 2$, $\Delta k = 411.76$; BAPs $k = 2$, $p = 0.96$), and Lahontan redside (STRUCTURE $k = 2$, $\Delta k = 304.53$; BAPs $k = 2$, $p = 0.979$)(Fig. 4), but not for Paiute sculpin, mountain whitefish, or Lahontan speckled dace (Fig. S1, Fig. 5).

To determine whether the population genetic structure observed for mountain sucker, Tahoe sucker, and Lahontan redside in 2008 was simply due to the larger sample sizes for 2008 compared to 2007, the 2008 data for each species was randomly subsampled using the 2007 sample sizes per section and re-ran the analysis. For each species 10 randomly sampled datasets were created and re-ran the Bayesian genotype clustering analysis (STRUCTURE) using a 10,000 burnin and five, 500,000 MCMC replicates per species. Although the number of genotype clusters observed in 2008, were not observed for all replicates for all species using 2007 numbers, the 2008 patterns of population genetic structure were observed for all species except Lahontan redside (Fig. S2, Table S4). Population genetic structure was observed using the 2007 sample sizes for Lahontan redside, but it did not follow the same pattern observed with the full 2008 dataset. This result is likely due to the lack of samples collected from the upper watershed sampling sites in 2007. Lahontan redside also had the greatest difference in number of sample sizes between 2007 ($N = 145$ individuals genotyped) and 2008 ($N = 509$). However, we conclude that the smaller sample sizes in 2007 did not account for the lack of population genetic structure observed in that year. Average relatedness among individuals per species per transect per year was also low (Fig. S3), therefore the population genetic structure observed in 2008 was not a result of sampling sibling groups within transects.

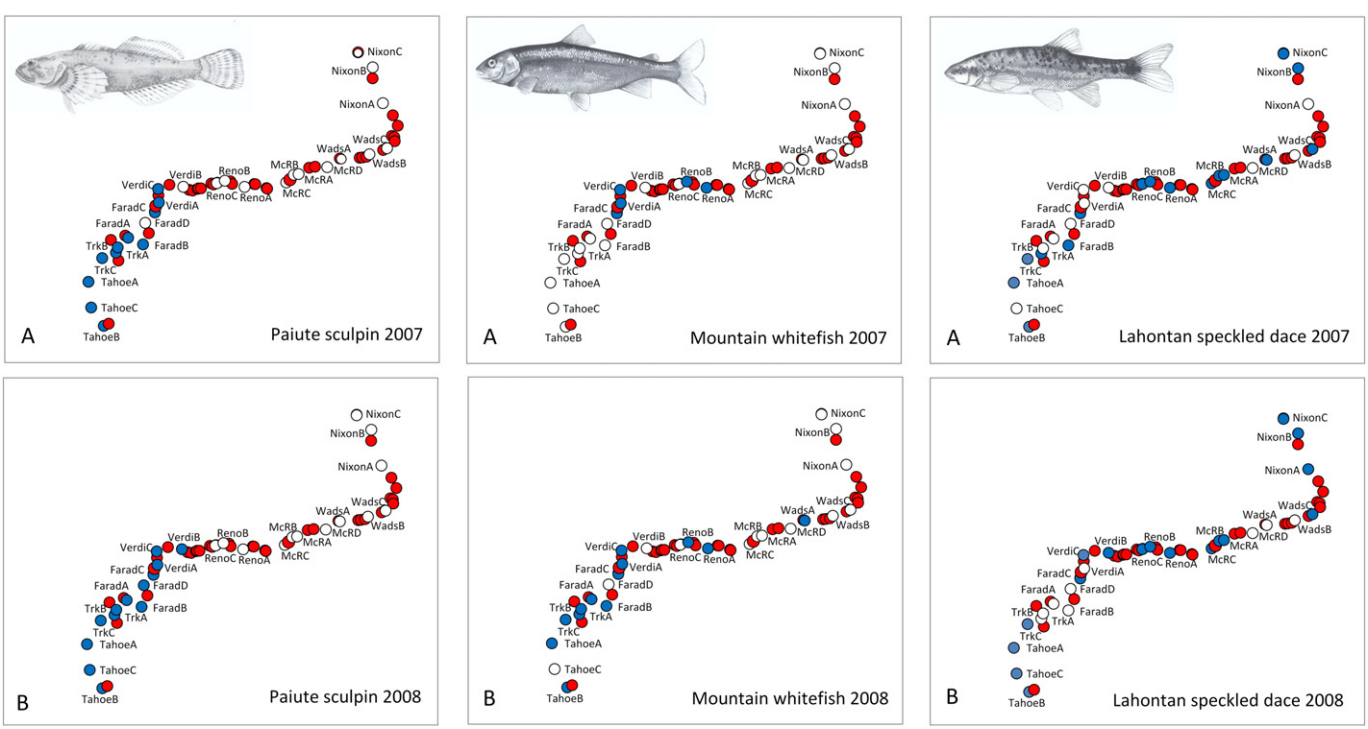
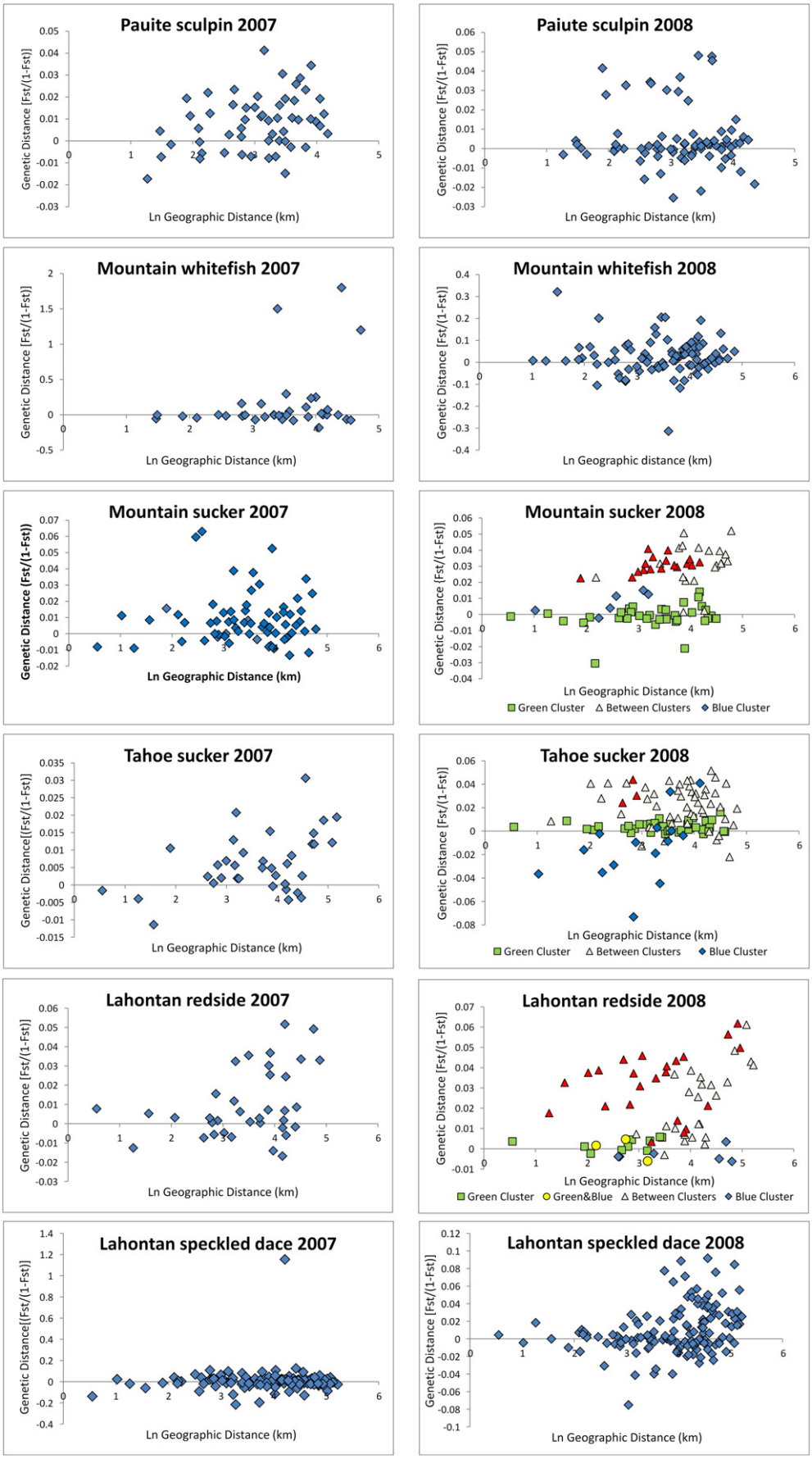


Fig. 5. Distribution of species with no population structure in either year of the study. Sampling sites (open or blue circles), and barriers (red circles) are graphed by latitude (Y axis) and longitude (X axis) to show location on the river. The name of each transect is indicated (see Table 1). Each panel represents a single species and year, (A, 2007; B 2008). All transects where a species was found are indicated by a blue circles.


Pauite sculpin 2007
Genetic Distance [Fst/(1-Fst)]
Ln Geographic Distance (km)
Paiute sculpin 2008
Genetic Distance [Fst/(1-Fst)]
Ln Geographic Distance (km)
Mountain whitefish 2007
Genetic Distance [Fst/(1-Fst)]
Ln Geographic Distance (km)
Mountain whitefish 2008
Genetic Distance [Fst/(1-Fst)]
Ln Geographic distance (km)
Mountain sucker 2007
Genetic Distance (Fst/(1-Fst))
Ln Geographic Distance (km)
Mountain sucker 2008
Genetic Distance [Fst/(1-Fst)]
Ln Geographic Distance (km)
Green Cluster
Between Clusters
Blue Cluster
Tahoe sucker 2007
Genetic Distance[(Fst/(1-Fst)]
Ln Geographic Distance (km)
Tahoe sucker 2008
Genetic Distance [Fst/(1-Fst)]
Ln Geographic Distance (km)
Green Cluster
Between Clusters
Blue Cluster
Lahontan redside 2007
Genetic Distance [Fst/(1-Fst)]
Ln Geographic Distance (km)
Lahontan redside 2008
Genetic Distance [Fst/(1-Fst)]
Ln Geographic Distance (km)
Green Cluster
Green&Blue
Between Clusters
Blue Cluster
Lahontan speckled dace 2007
Genetic Distance[(Fst/(1-Fst)]
Ln Geographic Distance (km)
Lahontan speckled dace 2008
Genetic Distance [Fst/(1-Fst)]
Ln Geographic Distance (km)

3.4.1. Catostomids

There were no significant pairwise F_{ST} between any transects for mountain suckers sampled in 2007 ($P = 0.0007$, obtained after 1320 permutations) results which are consistent with Bayesian genotype clustering results. In 2008, there were statistically significant pairwise F_{ST} between the VerdiB, RenoC, and RenoB (the three uppermost transects where mountain suckers were found, that were not differentiated from each other) and most transects lower in the river ($P = 0.0006$ obtained after 1560 permutations; Table S5a). There are three temporal and seasonal barriers (Table 1; Eastman, Glendale and Pioneer ditch diversions) between these three upper sites and the RenoA transect (the lower most Reno site). All individuals from the RenoA transect and nine transects down river assigned primarily to a single genotype cluster (green), while VerdiB, RenoC and RenoB assigned to a different genotype cluster (blue, Fig. 4). Only 162 individuals were sampled in the upper three transects where all genotyped individuals assigned to the blue cluster. Individual mountain suckers in the remaining nine transects (N = 199) occupied a large portion of the river and assigned either completely or primarily to the green genotype cluster (BAPs and STRUCTURE). An isolation-by-distance pattern was not observed in 2007 ($R^2 = 0.000$, $P = 0.949$) (Fig. 6). In 2008, transects within each genotype cluster and between cluster transect comparisons also did not show evidence of isolation-by-distance ($R^2 = 0.335, 0.022, 0.03$; $P = 0.134, 0.191, 0.156$ respectively) (Fig. 6). These results suggest that individuals from transects within each genotype cluster were able to mix.

Similarly, there were no significant pairwise F_{ST} between transects sampled in 2007 for the Tahoe sucker, however, the Mantel test showed a significant isolation-by-distance pattern ($R^2 = 0.189$, $P = 0.008$) suggesting limited movement by individuals. Transects RenoA and McRD were differentiated (pairwise F_{ST}) from most transects sampled below them in the river in 2008 ($P = 0.00047$, obtained after 2100 permutations; Table S5b). In 2008, there was a marginally significant isolation-by-distance pattern in the blue cluster ($R^2 = 0.198$, $P = 0.055$), which included the VerdiA to RenoA and McRD transects, but no pattern for either the green ($R^2 = 0.000$, $P = 0.797$), or comparisons between genotype clusters ($R^2 = 0.000$, $P = 0.38$). Sixty-three individuals were sampled in the VerdiA to RenoA and McRD transects while 942 were sampled in the remaining nine transects whose individuals all assigned to the green cluster. The break between the blue and green clusters comes between the RenoA and McRC transect, where Steamboat Creek flows into the main stem river, the hypothesized water quality barrier. Between the McRD transect, which assigns to the blue cluster, and WadsA in the green cluster is Derby Dam, a complete in-stream barrier to upstream fish passage. Between McRD and the upstream transect McRA, which assigns to the green cluster is the Hill Ditch diversion, a temporal and seasonal barrier (Table 1, Fig. 4).

3.4.2. Cottids

There were no significant pairwise F_{ST} between any transects sampled in 2007 for Paiute sculpin (F_{ST} range, −0.0018–0.039). In 2008, the FaradA transect was differentiated from three upstream transects; TahoeC ($F_{ST} = 0.0461$), TahoeB ($F_{ST} = 0.045$), and TrkC ($F_{ST} = 0.033$) ($P = 0.0006$, obtained after 1560 permutations). There was a significant isolation-by-distance pattern in 2007 ($R^2 = 0.084$, $P = 0.018$), but not 2008 ($R^2 = 0.000$, $P = 0.573$; Fig. 6). However, geographic distance explains very little of the variance in genetic distance in 2007. No genetic structure was observed despite three complete in-stream barriers and one temporal and seasonal barrier within the portion of the river occupied by Paiute sculpin (Figs. 3 and 5). However, the majority of Paiute sculpin occurred in transects upstream of the complete in-stream barriers in both years (upstream 2007, $N = 110$; 2008, $N = 319$: downstream 2007, $N = 44$; 2008, $N = 126$).

3.4.3. Cyprinids

There were no significant pairwise F_{ST} in 2007 for Lahontan redside transects, however, there was a significant isolation-by-distance pattern ($R^2 = 0.103$, $P = 0.032$), but the R^2 value was low and therefore geographic distance does not explain much of the observed variance in genetic distance (Fig. 6). In 2008, there were multiple genotype clusters and significant differentiation among a large number of transects ($P = 0.0007$, obtained after 1320 permutations; Table S5c, Fig. 4). All significant pairwise F_{ST} were between transects which assigned to different genotype clusters (Fig. 4). Both the STRUCTURE and BAPs analyses group the three most upstream occupied transects (TahoeB, RenoA, and McRC) together with spatially separated McRD into a single genotype cluster (blue). Only fifty-nine individuals were sampled in these three transects. Individual Lahontan redsides in the remaining eight transects (N = 957) occupied a large portion of the river and assigned either completely to the green genotype cluster (BAPs) or to a mixture of both green and blue (STRUCTURE) (Fig. 5). The break in genotype cluster assignment occurred between the McRC and McRB transects, which are separated by the McCarran Ditch diversion that is not considered a barrier, and again between McRD and McRA (upstream) and WadsA (downstream) which are separated by Hill Ditch diversion (temporal and seasonal barrier) and Derby Dam (complete in-stream barrier) respectively (Fig. 4). There was no pattern of isolation-by-distance among transects within or between genotype clusters ($R^2 \leq 0.000$, $P \geq 0.169$; Fig. 6).

In 2007, Lahontan speckled dace in the NixonB transect in the lower river were differentiated from FaradC ($F_{ST} = 0.039$) in the upper river and from the McRC mid-river ($F_{ST} = 0.02$), just below the Reno metropolitan area ($P = 0.0004$, obtained after 2400 permutations). There was no isolation-by-distance pattern observed in 2007 ($R^2 = 0$, $P = 0.525$) and while there was significant isolation-by-distance ($R^2 = 0.078$, $P = 0.001$) in 2008, very little of the variance in genetic distance was explained by geographic distance (Fig. 6). Only the TahoeB transect in the headwaters was differentiated (significant F_{ST}) from multiple downstream sites (Table S5d).

3.4.4. Salmonids

Despite the fact that the pairwise F_{ST} were typically larger than those observed for the other species (F_{ST} range, −0.1169–0.165), there were no statistically significant pairwise F_{ST} between any of the mountain whitefish transects in either year of the study (2007, $P = 0.001$, obtained after 720 permutations; 2008, $P = 0.0005$, obtained after 1820 permutations). An isolation-by-distance pattern was also not observed in either year ($R^2 = 0.05, 0$; $P = 0.102, 0.785$ respectively; Fig. 6).

3.5. Effective population size

Paiute sculpin had a comparatively high Ne compared to all other species in both years despite the fact that they occupied only 78 km of river habitat, the lowest of all species sampled (Table 4). However, the section of the river occupied by Paiute sculpin also had the fewest in-stream structures. Tahoe sucker, found primarily in transects within and below the Reno metropolitan area, occupied 123–176 km of habitat depending on the sampling year, had the next highest Ne both in 2007, when no genetic structure was evident, and 2008 when two genotype

Fig. 6. Mantel tests. Each panel represents a species and year. Genetic distance [$F_{ST}/(1 - F_{ST})$] (Y axis) and Ln geographic distance (X axis) were calculated between transects. In 2007 there was no population genetic structure for any species and the blue diamonds represent all pairwise comparisons. In 2008, for those species with population genetic structure (mountain sucker, Tahoe sucker, and Lahontan redside), tests for isolation-by-distance patterns were conducted for all pairwise comparisons within genotype clusters (transects within the green cluster = green squares; within the blue cluster = blue diamonds) and between clusters (transects between clusters = triangles). The red symbols represent significant pairwise F_{ST} between pairs of transects. Note that significant F_{ST}s were between transects whose individuals assigned to separate genotype clusters.

Table 4
Effective population size (N_e) per year, per species and genotype cluster with the 95% confidence interval, and kilometers (km) of river occupied.

	Paiute sculpin		Mountain whitefish		Lahontan speckled dace	
	2007	2008	2007	2008	2007	2008
Ne	996.8	3105.7	No estimate	1708	473.5	499.4
95%CI	590–2888.3	1161.8–8909.8	No estimate	268.2–infinity	322.1–858.5	417.7–615.5
River km occupied	65.34	78.03	111.82	128.85	184.79	184.79

	Mountain sucker			Tahoe sucker			Lahontan redside		
	2007	2008		2007	2008		2007	2008	
		Blue	Green		Blue	Green		Blue	Green
Ne	319	394.5	433.2	843.1	553.9	2012.8	194.6	83.9	177.5
95%CI	244.9–448.6	278.5–657.1	349–563.8	584.5–1467.9	316.6–1920.7	1047.8–18,087.3	167–231.3	75.8–93.6	167.2–188.6
River km occupied	127.75	32.22	89.89	176.51	95.40	76.51	131.30	108.78	73.47

clusters were identified (Table 4). Lahontan speckled dace and mountain sucker had lower Ne values ranging from 319 to 499. Lahontan redside had very low Ne in both years (Table 4). We were unable to calculate Ne for Mountain whitefish in 2007, but Ne estimated for 2008 was large (1708). However, we were unable to fit reasonable confidence interval around the 2008 estimate.

4. Discussion

The Truckee River is a medium sized river found in the interior Lahontan hydrographic basin of northern Nevada, northeastern California and southeastern Oregon. Flowing from an oligotrophic source to an endorheic terminus, the Truckee River supports a suite of fishes native to western United States waters. Some species found in the Truckee River, such as the Lahontan redside and Tahoe sucker, are narrowly distributed within Nevada and eastern California. As in most river drainages there was overlap in species distributions within the Truckee River, but there were also clear patterns of spatial segregation moving down the watershed with occurrence and abundance of individual species related to stream flow, temperature, gradient, elevation, substrate, and riparian vegetation which dictate allochthonous inputs and in-stream autochthonous production. In-stream structures varied from dams deemed complete barriers to upstream fish passage to structures classified as seasonal or temporal barriers to fish movement. In this study we observed spatial and temporal variability in the effectiveness of in-stream structures as barriers to movement for a suite of mostly small (except for the larger salmonid, the Mountain whitefish), short lived fish species, as a result of variation in total yearly discharge (see Fig. 2).

In 2007 and 2008 discharge into the main stem river below tributary reservoir inputs was 20–50% of the 2006 flows depending upon the USGS gauging station. We hypothesized that as a result of the high flows in 2006 movement over in-stream structures would be facilitated and we did not observe population genetic structure in 2007 for any species. The lower discharge in 2007 and 2008, however, set up a scenario whereby multiple in-stream structures were again effective at preventing movement of individual fish and population substructure was observed for three species, mountain and Tahoe suckers and Lahontan redside in 2008. Because population substructure is based ultimately upon differences in allele frequencies among groups, our results suggest fish with different subsets of alleles were trapped above and below specific in-stream structures as flows decreased. Distance between the in-stream structures varies and therefore the number of fish that subsections can support will also vary. If low levels of water isolate fish between barriers for long periods of time, random genetic drift will erode genetic variation and increase genetic differentiation.

We recognize that temperature and flow conditions are correlated and therefore that temperature likely plays a role in movement dynamics, an effect that we are unable to tease apart in this study. However, a PCA analysis of fish and environmental data collected from fall 2008 through 2010 at the same sites used in this study showed that river section ($r = 0.96$) and elevation ($r = 0.95$) explained most of the variance in fish distribution and density. River section and elevation had in turn negative associations with temperature (-0.68) and conductivity not surprisingly ($r = -0.89$; Moore, 2011). Additionally, fish distribution and density were highly correlated with flow ($r = 0.82$) and season ($r = -0.76$) suggesting that the biggest seasonal influence on the Truckee River was river discharge followed by the influence of discharge on river temperatures ($r = 0.50$; Moore, 2011). There were no sharp demarcations in temperature above and below the in-stream structures which separated genotype clusters (Moore, 2011). Thus, the observation of distinct genotype clusters above and below specific in-stream structures strongly suggests that low flow conditions in 2008 prevented upstream movement over these structures.

The top most complete barrier, Tahoe Dam, is at the outflow of the Truckee River from Lake Tahoe, and is used to manage lake level. This dam prevents fish from moving into Lake Tahoe from the river. The bottom most complete barrier, Marble Bluff Dam, prevents fish moving into the river from Pyramid Lake. USFWS operates a fish passage facility (essentially an elevator) at Marble Bluff to allow cui-ui and Lahontan cutthroat trout, ESA listed species, to pass into the lower river from Pyramid Lake for spawning. Paiute sculpin, mountain whitefish and Lahontan speckled dace have distributions which overlap the five upper most complete in-stream barriers. However, we did not observe population substructure for these species in either year of the study. For Paiute sculpin the upstream barriers were clustered at their lower distributional limit and downstream movement over these barriers was possible. There were no in-stream structures above these dams where the majority of sculpin were found. Additionally, no isolation-by-distance pattern within their distribution suggests Paiute sculpin movement was primarily unrestricted throughout the upper 78 km of river. Although mountain whitefish were found in transects above and below the five complete in-stream barriers found in the upper watershed, the majority of individuals were found in the Verdi (B and C) and Reno transects where multiple temporal or seasonal barriers occur, but movement dynamics of whitefish did not appear to be impacted by these in-stream structures. Lahontan speckled dace was sparsely distributed throughout the entire river, but found in greatest numbers in the NixonB and C transects below Numana Dam in the lower river.

The species with population substructure in 2008, mountain and Tahoe suckers and Lahontan redsides, were found primarily below the five complete in-stream barriers in the upper watershed, but overlapped with 12 seasonal or temporal barriers, one putative water quality barrier and several complete in-stream barriers. Below the creek input to the river, total mercury, nitrogen, phosphorus, and sulfate loads increase to three to four times the concentrations above the creek input (Pizarro-Barraza et al., 2014). Previous work, using mercury and isotopes, revealed limited upstream movement over the creek inflow point for all species examined suggesting this was a water quality

barrier (Pizarro-Barraza et al., 2014). The patterns of population substructure that emerged in 2008 for mountain and Tahoe sucker and Lahontan redside coincide primarily with in-stream structures, but not all species were similarly impacted by the same barriers.

There are three seasonal or temporal barriers separating the two genotype clusters identified for mountain suckers. These same structures, however, do not represent barriers to movement for Tahoe suckers and Lahontan redsides whose distributions also overlap with them. The change in genotype cluster membership for Tahoe suckers comes between the RenoA-McRC transects, which are separated by the water quality barrier discussed above, which does not appear to impact mountain suckers or Lahontan redsides, suggesting Tahoe suckers may be more sensitive to nutrient pollution. Lahontan redside individuals in adjacent transects (McRC and McRB) assign to different genotype clusters despite the fact that the McCarran ditch diversion, which separates these transects, is not considered a barrier. Tahoe suckers and Lahontan redsides individuals sampled in the McRD transect assign to the upstream blue clusters, which are disjunct from the McRD transect. One temporal or seasonal barrier separates McRD from McRA transect upstream and a complete in-stream barrier (Derby dam) separates McRD from WadsA downstream.

Although we cannot definitely say whether there was upstream movement, as fish floy tagged in this study were not recaptured, the fact that no population genetic structure was observed in 2007 suggests that fish can move at least downstream over in-stream structures in high water years. AMOVA results for all species showed genetic variation was primarily within individuals and most species did not have isolation-by-distance patterns in 2007, results which suggest widespread mixing and transitory population genetic structure due to barriers during differing flow conditions. Overall, these results suggest that species distributions within a watershed, which are ultimately determined by habitat preference and life history, together with barrier type and placement, and flow dynamics determine whether population substructure will emerge during low flow conditions.

The levels of genetic diversity observed for these fishes was quite high (see Table 4) and similar to, or even higher, than what has been reported for closely related species (Estoup et al., 1993; Scribner et al., 1996; Englbrecht et al., 1999; Sakamoto et al., 2000; Tranah et al., 2001; Baerwald and May, 2004; Rogers et al., 2004; Nolte et al., 2005; Girard and Angers, 2006; Cardall et al., 2007; Dubut et al., 2009; Robinson et al., 2009; Lamphere and Blum, 2012). Mountain whitefish had the overall lowest levels of allelic variation and heterozygosity, but was the least abundant species in the river and levels of genetic diversity were similar to what has been reported in the literature (Whiteley et al., 2004; Whiteley et al., 2006). Movement by mobile individuals and genetic mixing during periods of high flow has likely ameliorated the impact of in-stream barriers on the erosion of genetic diversity.

However, Ne was variable and tended to be smaller for species that exhibited population substructure. We recognize that the species in this study, albeit short lived, have overlapping generations and that we have estimated as per Luikart et al. (2010) something between Nb (number of breeding adults) and Ne and as a result may have underestimated Ne. However, our aim is to compare Ne among species in the context of emergent population structure and not necessarily to estimate a precise Ne per species. We interpret the low Ne as consistent with long term population fluctuations driven by subpopulation isolation between barriers, losses of genetic variation due to bottlenecks, small subpopulations, and random genetic drift.

Although we only observed the emergence of two genotype clusters for those species with population genetic structure in 2008 we have only two years of data. The distribution of these species in the river is interspersed with 12 separate temporal and seasonal barriers and additional subpopulation structure is likely under continued dry conditions as we have seen in the western United States where recent snow packs have been well below average.

Habitat fragmentation has greatly impacted a wide variety of taxa – e.g., the Mediterranean esparto grasshopper (*Ramburiella hispanica*) (Ortego et al., 2015), golden lion tamarin (*Leontopithecus rosalia*) (Zeigler et al., 2013), Mojave tortoise (*Gopherus agassizii*) (Averill-Murray et al., 2013), mountain lion (*Puma concolor*) (Foley et al., 2013), yellow-tufted (*Lichenostomus melanops*) and white-plumed (*Lichenostomus penicillatus*) honeyeaters (Harrisson et al., 2014) – and not least of which are the fresh water fishes (Labonne and Gaudin, 2006; Peterson et al., 2008; Sato and Harada, 2008; Nislow et al., 2011; Pepino et al., 2012). Much research has focused on apex salmonid predators and the impact that fragmentation of once large, multiple order stream networks has had on population viability (Neville et al., 2006; Gresswell and Hendricks, 2007; Fausch et al., 2009; Sanderson and Hubert, 2009; Alexiades et al., 2012; Pepino et al., 2012). However, declines in population numbers due to water diversions and habitat fragmentation, for many small native species found in inland waters, has received much less attention (but see Roberts et al., 2013; Osborne et al., 2014). In-stream structures which can isolate larger species may have even greater impacts to the long term population viability of smaller less mobile species that have more restricted distributions, smaller effective population sizes, and a greater dependence upon gene flow to maintain genetic diversity. In a mark-recapture study on the endangered Bullhead (*Cottus gobio*) most marked individuals moved <10 m from their capture location, with 35% of recaptured individuals moving up to 270 m from their capture site (Knaepkens et al., 2005), suggesting limited movement and fine scale spatial structure for this sculpin. In contrast, a mark-recapture study on Potomac sculpin (*Cottus girardi*) showed that while the majority of marked individuals stayed within 30 m of their original capture location some individuals moved as far as 2 km (Hudy and Shiflet, 2009). Genetic structure has been found among catchments for the prairie stream fish, the creek chub (*Semotilus atromaculatus*), which was attributed to anthropogenic impoundments (reservoirs) with further substructure within catchment due to habitat types (Hudman and Gido, 2013). Lentic habitat represents a barrier to movement for this species, thereby reducing the probability of genetic rescue (gene flow and introduction of additional allelic variation) for populations in tributaries that drain into impoundments (Hudman and Gido, 2013). Other examples include the small and endangered benthic Roanke logperch (*Percina rex*) where hydroelectric dam placement explains the pattern of genetic structure among streams within a catchment basin over other hypotheses such as isolation-by-distance, with some isolated populations having less than half the allelic diversity of populations found in larger stream fragments (Roberts et al., 2013).

In studies of multiple species found in the same watershed, the impact of barriers on individual species varies, not surprisingly. Hanging culverts were a complete upstream barrier to native burbot (*Lota lota*), and a partial impediment to movement for native spoonhead sculpin (*Cottus ricei*), sucker and minnow species, but did not prevent upstream movement of nonnative rainbow trout (*Oncorhynchus mykiss*) (MacPherson et al., 2012). Similarly, brook trout (*Salvelinus fontinalis*) were able to move upstream over culvert barriers, but many smaller species, e.g., catostomids (sucker spp.) and cyprinids (chub and dace spp.) were not (Nislow et al., 2011). While, three native catostomid spp. (bluehead suckers *Catostomus discobolus*, flannelmouth suckers *Catostomus latipinnis*, and roundtail chub *Gila robusta*) were able to move both upstream and downstream over an irrigation diversion dam, they could not move over a fish barrier designed to exclude nonnative salmonids from the upper watershed (Compton et al., 2008). Barrier type is clearly an important determinant of fish passage.

The genetic results of this study show that in-stream structures found in the main stem Truckee River can fragment riverine habitat isolating fish between barriers, which depending upon flow conditions, can isolate fish for varying periods of time. Longer periods of isolation between barriers due to low flow conditions can thus lead to reductions in genetic variation and impact long term population viability. The

highly variable yearly discharge in this system is driven largely by drought cycles and global climate change is likely to exacerbate frequency and duration of population structure as well as changes in timing, amount, and type of precipitation. However, management of stream discharge could ameliorate at least temporally the impacts of in-stream structures found in many river systems especially for species with both sedentary and mobile individuals, which can move considerable distances, thereby providing connectivity among groups of sedentary fish within a watershed.

5. Conclusions

The results of this study illustrate the complexities of water management for large riverine ecosystems, heavily impacted by anthropogenic use, that support a wide variety of fishes and have a large number of non-randomly spaced in-stream structures of various types. Removal of fish passage barriers in recent years has largely focused on in-stream structures deemed complete barriers (Stanley et al., 2007; Kornis et al., 2015). However, the majority of species in this study were not affected by such barriers due lack of spatial overlap with many of these structures, but were impacted by the numerous temporal and seasonal barriers, such as water diversions and pollution inputs, barrier types which have received much less attention in addressing fish passage issues. Given the large number of dams and diversions on the Truckee River, dam mitigation or removal may not be feasible or socially palatable in the short term. A more "ecosystem based" approach, however, could include strategic management of water releases from in-stream reservoirs in order to mimic the natural hydrograph as well as the use of periodic pulse flows to overcome the temporal and seasonal barriers to movement for the suite of small native fishes and maintain connectivity. Such a flow management approach could reduce the impacts of small population size, mitigate genetic bottlenecks, and decrease losses of genetic variation through random genetic drift over periods of prolonged drought. Indeed, Leasure et al. (2016) characterized natural flow regimes in the Ozark-Ouachita interior highlands region of Arkansas in order to identify ecohydrologic regions potentially useful for conservation planning for communities of aquatic organisms.

We have shown with this work that in-stream structures and "water quality" barriers together with flow dynamics can influence whether and when these features function as barriers to fish movement. For many streams and rivers that were once free flowing, multiple in-stream structures now overlap with a wide variety of fish communities. Even structures deemed temporal or seasonal can have an impact on movement and genetic mixing. Global climate change may influence the duration of low flow periods, emergence of subpopulations and extent of isolation. The take home conservation message here is that in-stream structures need to be evaluated in the context of the entire fish community. While whole scale removal of structures within rivers is unfeasible, understanding of the impacts in-stream structures and water quality changes to species with different life histories is the first step in formulating management strategies.

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